then

$$P(E_1 \text{ or } E_2) = P(E_1 \cup E_2) = P(E_1) + P(E_2) - P(E_1 \cap E_2)$$

① one card is drawn from a pack of 52 cards. What is the probability that is either a spade (or) an ace.

Let,

'A' ~~be the an event of a spade~~

Let, 'S' be the sample space of all sample events is

$$n(S) = 52$$

'A' be the probability of spade is $\Rightarrow P(A) = \frac{n(A)}{n(S)} = \frac{13}{52}$

Let, 'B' be the probability of an ace is $\Rightarrow P(B) = \frac{n(B)}{n(S)} = \frac{4}{52}$

Let, the Probability of a spade and an ace is $\Rightarrow P(A \cap B) = \frac{1}{52}$

The Probability of a spade (or) ace is.

By using addition Theorem,

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$= \frac{13}{52} + \frac{4}{52} - \frac{1}{52}$$

$$= \frac{13+4-1}{52}$$

$$= \frac{17-1}{52}$$

$$= \frac{\cancel{16}\,4}{\cancel{52}\,13} = \frac{4}{13}$$

② The chance of three students A, B, C solving a problem given in mathematics olympiad are $\frac{1}{2}$, $\frac{3}{4}$, and $\frac{1}{4}$ respectively. What is the probability that the problem is solved.

Given that,

The Probability of 'A' student $\Rightarrow P(A) = \frac{1}{2}$

The Probability of 'B' student $\Rightarrow P(B) = \frac{3}{4}$

The Probability of 'C' student $\Rightarrow P(C) = \frac{1}{4}$

we know that,

$$P(\bar{A}) = 1 - P(A) = 1 - \frac{1}{2} = \frac{1}{2}$$

$$P(\bar{B}) = 1 - P(B) = 1 - \frac{3}{4} = \frac{1}{4}$$

$$P(\bar{C}) = 1 - P(C) = 1 - \frac{1}{4} = \frac{3}{4}$$

The Probability of that the problem is not solved is

$$P(\bar{A} \cap \bar{B} \cap \bar{C}) = P(\bar{A}) \cdot P(\bar{B}) \cdot P(\bar{C})$$

$$= \frac{1}{2} \cdot \frac{1}{4} \cdot \frac{3}{4} = \frac{3}{32}$$

The Probability of that the problem is solved is

$$P(A \cup B \cup C) = 1 - P(\bar{A} \cap \bar{B} \cap \bar{C})$$

$$= 1 - \frac{3}{32}$$

$$= \frac{32-3}{32}$$

$$= \frac{29}{32}$$

$P(A) = P(E_1)\,P\left[\frac{A}{E_1}\right] + P(E_2)\,P\left[\frac{A}{E_2}\right] + \cdots + P(E_n)\,P\left[\frac{A}{E_n}\right]$

① In a group consisting of equal number of men and women 10% of men and 45% of the women are unemployed. If a person is selected for randomly from the group, then find the probability that the person is an employee.

Let 'S' be the sample space of all events is $S = \{M, W\}$

Let,

$P(M)$ = The probability of Men $= \frac{n(M)}{n(S)} = \frac{1}{2}$

$P(W)$ = The probability of Women $= \frac{n(W)}{n(S)} = \frac{1}{2}$

Let, 'A' be denotes unemployed
'B' be denotes employed

Now,

$P\left(\frac{A}{M}\right) = 10\% = \frac{10}{100}$

$P\left(\frac{A}{M}\right) = \frac{1}{10}$

$P\left(\frac{A}{W}\right) = 45\%$

$= \frac{45}{100}$

$P\left(\frac{E}{M}\right) = 1 - \left(P\left(\frac{E}{M}\right)\right)'$

$= 1 - P\left(\frac{A}{M}\right)$

$= 1 - \frac{1}{10}$

$= \frac{9}{10}$

$P\left(\frac{E}{W}\right) = 1 - \left(P\left(\frac{E}{W}\right)\right)'$

$= 1 - P\left(\frac{A}{W}\right)$

$= 1 - \frac{45}{100}$

$= \frac{55}{100}$

The probability that the person is an unemployed is

$P(E) = P(M) \cdot P\left(\frac{E}{M}\right) + P(W) \cdot P\left(\frac{E}{W}\right)$

$= \frac{1}{2}\left(\frac{9}{10}\right) + \frac{1}{2}\left(\frac{55}{100}\right)$

$= \frac{1}{2}\left[0.9 + 0.55\right]$

$= \frac{1}{2}\left[\cancel{1.45}\right]$ 0.725

$\boxed{P(E) = 0.725}$

② If A, B are two events such that $P(A) = \frac{1}{3}$, $P(B) = \frac{3}{4}$ and $P(A \cup B) = \frac{11}{12}$. Then find $P\left(\frac{A}{B}\right)$ and $P\left(\frac{B}{A}\right)$

$\boxed{P(E) = 0.725}$

(2) If A, B are two events such that $P(A) = \frac{1}{3}$, $P(B) = \frac{3}{4}$ and $P(A \cup B) = \frac{11}{12}$. Then find $P\left(\frac{A}{B}\right)$ and $P\left(\frac{B}{A}\right)$

Given,

$P(A) = \frac{1}{3}$

$P(B) = \frac{3}{4}$

$P(A \cup B) = \frac{11}{12}$

By using Addition theorem,

$P(A \cup B) = P(A) + P(B) - P(A \cap B)$

$$P(A \cap B) = P(A) + P(B) - P(A \cup B)$$

$$P(A \cap B) = \frac{1}{3} + \frac{3}{4} - \frac{11}{2}$$

$$P(A \cap B) = \frac{4+9-11}{12} = \frac{2}{12} = \frac{1}{6}$$

Now,

$$P(A/B) = \frac{P(A \cap B)}{P(B)}$$

$$= \frac{1/6}{3/4}$$

$$= \frac{1}{\cancel{6}_{3}} \times \frac{\cancel{4}^{2}}{3}$$

$$P\left(\frac{A}{B}\right) = \frac{2}{9}$$

$$P\left(\frac{B}{A}\right) = \frac{P(A \cap B)}{P(A)}$$

$$= \frac{1/6}{1/3}$$

$$= \frac{1}{\cancel{6}_{2}} \times \cancel{3}$$

$$P\left(\frac{B}{A}\right) = \frac{1}{2}$$

$P(A) = \frac{2}{\ldots}$, $P(B) = \frac{9}{12}$, $P(A \cap B) = \frac{4}{13}$ find $P\left(\frac{A}{B}\right)$.

$$P\left(\frac{E_K}{A}\right) = \frac{P(E_K) \cdot P(A/E_K)}{P(E_1)\,P\left(\frac{A}{E_1}\right) + P(E_2)\,P\left(\frac{A}{E_2}\right) + \cdots + P(E_n)\,P\left(\frac{A}{E_n}\right)}$$

① In a bolt machine factory, 'A', B & C manufacture respectively 25%, 35% and 40% of the total of their output 5, 4, 2 percents are defective bolts. 'A' bolt is drawn at random from the product and is found to be defective. What are the probabilities that it was manufactured machines 'A', B & C.

Let,

'A' be the Event manufactured by first Machine

'B' be the Event manufactured by Second Machine

'C' be the Event manufactured by Third Machine

Given,

$$P(A) = 25\% = \frac{25}{100}$$

$$P(B) = 35\% = \frac{35}{100}$$

$$P(C) = 40\% = \frac{40}{100}$$

Let, 'D' denotes that the defective, Probability of A, B, C is

$$P\left(\frac{D}{A}\right) = 5\% = \frac{5}{100}$$

$$P\left(\frac{D}{B}\right) = 4\% = \frac{4}{100}$$

$$P\left(\frac{D}{C}\right) = 2\% = \frac{2}{100}$$

i) If bolt is defective, then the probability that it from machine 'A'

$$P\left(\frac{A}{D}\right) = \frac{P(A).P\left(\frac{D}{A}\right)}{P(A).P\left(\frac{D}{A}\right) + P(B)\,P\left(\frac{D}{B}\right) + P(C).P\left(\frac{D}{C}\right)}$$

$$= \frac{\left(\frac{25}{100}\right)\left(\frac{5}{100}\right)}{\left(\frac{25}{100}\right)\left(\frac{5}{100}\right) + \left(\frac{35}{100}\right)\left(\frac{4}{100}\right) + \left(\frac{40}{100}\right)\left(\frac{2}{100}\right)}$$

$$= \frac{125/\cancel{(100)^2}}{\frac{125+140+80}{\cancel{(100)^2}}}$$

$$= \frac{\cancel{125}\ 25}{\cancel{345}\ 69} = \frac{25}{69}$$

ii) If bolt is defective, then the probability that it from machine B

$$P\left(\frac{B}{D}\right) = \frac{P(B).P\left(\frac{D}{B}\right)}{P(A).P\left(\frac{D}{A}\right) + P(B)\,P\left(\frac{D}{B}\right) + P(C).P\left(\frac{D}{C}\right)}$$

$$P\left(\frac{B}{D}\right) = \frac{\left(\frac{35}{100}\right)\left(\frac{4}{100}\right)}{345/(100)^2}$$

$$= \frac{140/\cancel{(100)^2}}{345/\cancel{(100)^2}} = \frac{\cancel{140}\ 28}{\cancel{345}\ 69} = \frac{28}{69}$$

iii) If bolt is defective, then the probability that it from machine 'C'

$$P\left(\frac{C}{D}\right) = \frac{P(C).P\left(\frac{D}{C}\right)}{P(A).P\left(\frac{D}{A}\right) + P(B)\,P\left(\frac{D}{B}\right) + P(C).P\left(\frac{D}{C}\right)}$$

$$= \frac{\left(\frac{40}{100}\right)\left(\frac{2}{100}\right)}{\frac{345}{(100)^2}}$$

$$= \frac{80/(100)^2}{345/(100)^2}$$

$$= \frac{\cancel{80}\ 16}{\cancel{345}\ 69} = \frac{16}{69}$$

② In a bolt manufacturing company 3 machine man
[illegible] 30% 50% of its total output and of

$$\sigma^2 = E(x^2) - [E(x)]^2$$

Continous Distribution Function of Variance σ^2

$$\sigma^2 = \int_{-\infty}^{\infty} x^2 f(x)\,dx - \mu^2$$

A Random Variable 'X' has the following probability function

x	0	1	2	3	4	5	6	7
$P(x)$	0	K	2K	2K	3K	K^2	$2K^2$	$7K^2+K$

i) Find the Value of 'K'

ii) Evaluate $P(x<6)$, $P(x>6)$

iii) Evaluate $P(0<x<5)$

i) The total Probability $=1$.

i.e $\sum_{i=0}^{7} P(X=x_i) = 1$

$P(X=0)+P(X=1)+P(X=2)+P(X=3)+P(X=4)+P(X=5)+P(X=6)+P(X=7)=1$

$0+K+2K+2K+3K+K^2+2K^2+7K^2+K=1$

$9K+10K^2=1$

$10K^2+10K-K-1=0$

$10K(K+1)-1(K+1)=0$

$(10K-1)\cdot(K+1)=0$

$10K-1=0$

$10K=1$

$\boxed{K=\frac{1}{10}}$

$K+1=0$

$K=-1$ $[P(5)\geq 0]$

ii) $P(X<6) = P(X=0)+P(X=1)+P(X=2)+P(X=3)+P(X=4)+P(X=5)$

$= 0+K+2K+2K+3K+K^2$

$= 8K+K^2$

$= 8\left(\frac{1}{10}\right)+\frac{1}{100}$

$= \frac{80+1}{100} = \frac{81}{100} = 0.81$

$P(X>6) = P(X=7)$

$= 7K^2 + K$

$= 7\left(\frac{1}{100}\right) + \frac{1}{10} = \frac{7+10}{100} = \frac{17}{100} = 0.17$

iii) $P(0<X<5) = P(X=1) + P(X=2) + P(X=3) + P(X=4)$

$= K + 2K + 2K + 3K$

$= 8K = 8\left(\frac{1}{10}\right) = 0.8$

(2) The Probability Density function of a variant 'X' is

The Minimum value of $K = \frac{1}{30}$

③ If the probability Density Function of a random variable is given by

$$f(x) = \begin{cases} K(1-x^2), & \text{for } 0<x<1 \\ 0, & \text{otherwise} \end{cases}$$

Find the value of 'K' and the Probability that a random variable having this probability density will take on a value.

i) Between 0.1 & 0.2

ii) Greater than 0.5

Given function is,

$$f(x) = \begin{cases} K(1-x^2), & \text{for } 0<x<1 \\ 0, & \text{otherwise} \end{cases}$$

WKT,

$$\int_{-\infty}^{\infty} f(x)\,dx = 1$$

$$\int_{-\infty}^{0} f(x)\,dx + \int_{0}^{1} f(x)\,dx + \int_{1}^{\infty} f(x)\,dx = 1$$

$$0 + \int_0^1 f(x)\,dx + 0 = 1$$

$$\int_0^1 f(x)\,dx = 1$$

$$\int_0^1 k(1-x^2)\,dx = 1$$

$$k\left[x - \frac{x^3}{3}\right]_0^1 = 1$$

$$k\left[\left(1-\frac{1}{3}\right) - 0\right] = 1$$

$$k\left(\frac{2}{3}\right) = 1$$

$$\boxed{k = \frac{3}{2}}$$

(ii) The Probability that the varient will take on a value b/w 0.1 and 0.2 is

$$P(0.1 < x < 0.2) = \int_{0.1}^{0.2} f(x)\,dx$$

$$= \int_{0.1}^{0.2} k(1-x^2)\,dx$$

$$= k\left[x - \frac{x^3}{3}\right]_{0.1}^{0.2}$$

$$= \frac{3}{2}\left[\left(0.2 - \frac{(0.2)^3}{3}\right) - \left(0.1 - \frac{(0.1)^3}{3}\right)\right]$$

$$= \frac{3}{2}\left[0.2 - \frac{0.008}{3} - 0.1 + \frac{0.001}{2}\right]$$

$$= \frac{3}{2}\left[0.1 - \frac{0.007}{3}\right]$$

$$= \frac{\cancel{3}}{2}\left[\frac{0.3 - 0.007}{\cancel{3}}\right]$$

$$= \frac{1}{2}\left[0.293\right] = 0.1465$$

7) The Probability that the variant will take on a value greater than 0.5 is

$$P(x>0.5) = \int_{0.5}^{\infty} f(x)\,dx$$

$$= \int_{0.5}^{1} f(x)\,dx + \int_{1}^{\infty} f(x)\,dx$$

$$= \int_{0.5}^{1} f(x)\,dx + 0$$

$$= \int_{0.5}^{1} k(1-x^2)\,dx$$

$$= k\left(x - \frac{x^3}{3}\right)\Big]_{0.5}^{1}$$

$$= \frac{3}{2}\left[\left(1-\frac{1}{3}\right) - \left(0.5 - \frac{(0.5)^3}{3}\right)\right]$$

$$= \frac{3}{2}\left[\frac{2}{3} - 0.5 + \frac{0.125}{3}\right]$$

$$= \frac{\cancel{3}}{2}\left(\frac{2-1.5+0.125}{\cancel{3}}\right)$$

$$= \frac{2.125-1.5}{2} = \frac{0.625}{2} = 0.3125$$

) A Sample of 4 items is selected at random from a box containing 12 items of which f 5 are defective. find the expected number E. of defective items

$$= \frac{2.125 - 1.5}{2} = \frac{0.625}{2} = 0.3125$$

(4) A sample of 4 items is selected at random from a box containing 12 items of which 5 are defective. Find the expected number E. of defective items

Let, 'S' denotes the no of defective items among four items drawn from 12 items.

Clearly 'X' can take the values 0, 1, 2, 3, 4

No of good items = 7

No of defective items = 5

$P(x=0) = P(\text{no of defective items})$

$$= \frac{{}^7C_4 \, {}^5C_0}{{}^{12}C_4}$$

$$= \frac{\cancel{735}\,(1)}{\cancel{495}\,99}$$

$$= \frac{7}{99}$$

$${}^7C_4 = \frac{7\times6\times5\times4}{4\times3\times2\times1} = 35$$

$${}^{12}C_4 = \frac{12\times11\times10\times9}{4\times3\times2\times1}$$

$$= 55\times9$$

$P(x=1) = P(\text{no of defective and 3 good items})$

$$= \frac{{}^7C_3 \times {}^5C_1}{{}^{12}C_4} = \frac{\cancel{735}(5)}{\cancel{495}\,99} = \frac{35}{99}$$

$P(x=2) = P(\text{2 defective and 2 good items})$

$$= \frac{{}^7C_2 \times {}^5C_2}{{}^{12}C_4} = \frac{21\,(10)^2}{\cancel{495}\,99} = \frac{42}{99}$$

$P(x=3) = P(\text{3 defective and one good item})$

$$= \frac{{}^7C_1 \times {}^5C_3}{{}^{12}C_4} = \frac{7\,(10)^2}{\cancel{495}\,99} = \frac{14}{99}$$

$P(x=4) = P(\text{4 defective items and 0 good items})$

$$= \frac{{}^7C_0 \times {}^5C_4}{{}^{12}C_4} = \frac{1\,(5)\,1}{\cancel{495}\,99} = \frac{1}{99}$$

The discrete random variable is

$x = x_i$	0	1	2	3	4
$P(x = x_i)$	7/99	35/99	72/99	14/99	1/99

The expected value of defectives is

$$E[X] = \sum_{i=0}^{n} x_i P_i$$

$$= x_1 P(x=0) + x P(x=1) + x_2 P(x=2) + x_3 P(x=3) + x P(x=4)$$

$$= 0\,(7/99) + 1\,(35/99) + 2\,(42/99) + 3\,(14/99) + 4\,(1/99)$$

$$= 165/99 = 1.66$$

⑤ Find the mean and variance of the uniform probability distribution given by $f(n) = \frac{1}{n}$ for $x = 1, 2, --- n$.

Given,

$f(x) = \frac{1}{n}$ for $x = 1, 2, ---- n$

The probability distribution is

$x = x_i$	1	2	3	4	---	n
$P(X = x_i) = f(x_i)$	$1/n$	$1/n$	$1/n$	$1/n$	---	$1/n$

$$\text{Mean } (\mu) = \sum_{i=1}^{n} x_i p_i$$

$$= 1\left(\frac{1}{n}\right) + 2\left(\frac{1}{n}\right) + 3\left(\frac{1}{n}\right) + --- + n\left(\frac{1}{n}\right)$$

$$= \frac{1}{n}\left[1 + 2 + 3 + --- + n\right]$$

$$= \frac{1}{n}\left[\frac{n(n+1)}{2}\right]$$

$$\mu = \frac{n+1}{2}$$

$$\text{Variance } (\sigma^2) = \sum_{i=1}^{n} x_i^2 p_i - \mu^2$$

$$\sigma^2 = \left[(1)^2\left(\frac{1}{n}\right) + 2^2\left(\frac{1}{n}\right) + 3^2\left(\frac{1}{n}\right) + --- + n^2\left(\frac{1}{n}\right)\right] - \left(\frac{n+1}{2}\right)^2$$

$$= \frac{1}{n}\left[1^2 + 2^2 + 3^2 + --- + n^2\right] - \frac{(n+1)^2}{4}$$

$$= \frac{1}{n}\left[\frac{n(n+1)(2n+1)}{6}\right] - \frac{(n+1)^2}{4}$$

$$= \frac{(n+1)(2n+1)}{6} - \frac{(n+1)^2}{4}$$

$$= \frac{n+1}{2}\left(\frac{2n+1}{3} - \frac{n+1}{2}\right) = \frac{n+1}{2}\left[\frac{4n+2-3n-3}{6}\right]$$

$$= \frac{n+1}{12}(n-1) = \frac{n^2-1}{12}$$

① The Commulative distribution function for a continous Random variable 'X' is $F(x) = \begin{cases} 1-e^{-2x} & , x \geq 0 \\ 0 & , x < 0 \end{cases}$

Then find

i) Density Function

ii) Mean

iii) Variance

Given function,

$$f(x) = \begin{cases} 1-e^{-2x} & , x \geq 0 \\ 0 & , x < 0 \end{cases}$$

(i) The density function of $f(x)$ is

$$f(x) = \frac{d}{dx}\left[F(x)\right]$$

$$= \frac{d}{dx}\left[1-e^{-2x}\right]$$

$$= 0 - e^{-2x}(-2)$$

$$f(x) = 2e^{-2x}$$

$$f(x) = \begin{cases} 2e^{-2x} & , x \geq 0 \\ 0 & , x < 0 \end{cases}$$

(ii) Mean $\mu = E[x] = \int_{-\infty}^{\infty} x f(x)\, dx$

$$= \int_{-\infty}^{0} x f(x)\, dx + \int_{0}^{\infty} x f(x)\, dx$$

$$= 0 + \int_{0}^{\infty} x\left(2e^{-2x}\right) dx$$

$$= 2\int_{0}^{\infty} x e^{-2x}\, dx$$

$$\left[\because \int f(x) g(x)\, dx = f(x)\int g(x)\, dx - \int\left[f'(x)\int g(x)\, dx\right] dx + C\right]$$

ILATE Rule

Take,

$f(x) = x \qquad g(x) = e^{-2x}$

$f'(x) = 1 \qquad \int g(x)\,dx = \frac{e^{-2x}}{-2}$

$$= 2\left[x\frac{e^{-2x}}{-2} - \int 1\frac{e^{-2x}}{-2}dx\right]_0^\infty$$

$$= 2\left[-\frac{xe^{-2x}}{2} + \frac{1}{2}\int e^{-2x}dx\right]_0^\infty$$

$$= 2\left[-\frac{xe^{-2x}}{2} + \frac{1}{2}\left(\frac{e^{-2x}}{-2}\right)\right]_0^\infty$$

$$= \cancel{2}\left(\frac{-1}{\cancel{2}}\right)\left[xe^{-2x} + \frac{e^{-2x}}{2}\right]_0^\infty$$

$$= -\left[\frac{2xe^{-2x} + e^{-2x}}{2}\right]_0^\infty$$

$$= \frac{-1}{2}\left[e^{-2x}\cdot(2x+1)\right]_0^\infty$$

$$= \frac{-1}{2}\left[e^{-2(\infty)}\cdot(2(\infty)+1) - e^{0}(2(0)+1)\right]$$

$$= \frac{-1}{2}\left[0 - 1(0+1)\right]$$

$$= \frac{-1}{2}(-1)$$

$$\mu = \frac{1}{2}$$

(iii) Variance $(\sigma^2) = \int_{-\infty}^{\infty} x^2 f(x)\,dx - \mu^2$

$$= \int_{-\infty}^{0} x^2 f(x)\,dx + \int_{0}^{\infty} x^2 f(x)\,dx - \mu^2$$

$$= 0 + \int_{0}^{\infty} x^2 (2e^{-2x})\,dx - \left(\frac{1}{2}\right)^2$$

$$= 2\int_{0}^{\infty} x^2 e^{-2x}\,dx - \frac{1}{4}$$

Take,

$f(x) = x^2 \qquad g(x) = e^{-2x}$

$f'(x) = 2x \qquad g'(x) = \frac{e^{-2x}}{-2}$

$$\sigma^2 = 2\left[x^2\left(\frac{e^{-2x}}{-2}\right) - \int 2x\left(\frac{e^{-2x}}{-2}\right)dx\right]_0^{\infty} - \frac{1}{4}$$

$$= 2\left[\frac{-x^2 e^{-2x}}{2} + \int x e^{-2x}\,dx\right]_0^{\infty} - \frac{1}{4}$$

$$= 2\left[\frac{-x^2}{2}e^{-2x} + x\frac{e^{-2x}}{-2} - \int 1\frac{e^{-2x}}{-2}\,dx\right]_0^{\infty} - \frac{1}{4}$$

$$= 2\left[\frac{-x^2}{2}e^{-2x} - \frac{x}{2}e^{-2x} + \frac{1}{2}\left(\frac{e^{-2x}}{-2}\right)\right]_0^{\infty} - \frac{1}{4}$$

$$= \cancel{2}\left[-\frac{e^{-2x}}{\cancel{2}}\left(x^2 + x + \frac{1}{2}\right)\right]_0^{\infty} - \frac{1}{4}$$

$$= -e^{-\infty}\left[\infty^2 + \infty + \frac{1}{2}\right] + e^{0}\left[0 + 0 + \frac{1}{2}\right] - \frac{1}{4}$$

$$= 0 + \frac{1}{2} - \frac{1}{4}$$

$$= \frac{2-1}{4}$$

$$= \frac{1}{4}$$

$= \frac{10}{(9-5)!\,5!}\left(\frac{1}{2}\right)$

② 10 coins are thrown simultaneously. find the probability of getting atleast

i) 7 heads

ii) 6 heads

Given that,

$n = 10$

$p =$ probability of getting head $= \frac{1}{2}$

$q =$ probability of not getting head $= \frac{1}{2}$

By using binomial distribution, we have

$P(X=x) = {}^{n}C_{x}\, p^{x}\, q^{n-x}$

$P(X=x) = {}^{10}C_{x} \left(\frac{1}{2}\right)^{x} \left(\frac{1}{2}\right)^{10-x}$

(i) Atleast 7 heads

$$P(X \geq 7) = P(X=7) + P(X=8) + P(X=9) + P(X=10)$$

$$= {}^{10}C_7 \left(\frac{1}{2}\right)^7 \left(\frac{1}{2}\right)^{10-7} + {}^{10}C_8 \left(\frac{1}{2}\right)^8 \left(\frac{1}{2}\right)^{10-8} + {}^{10}C_9 \left(\frac{1}{2}\right)^9 \left(\frac{1}{2}\right)^{10-9} + {}^{10}C_{10} \left(\frac{1}{2}\right)^{10} \left(\frac{1}{2}\right)^{10-10}$$

$$= {}^{10}C_7 \left(\frac{1}{2}\right)^7 \left(\frac{1}{2}\right)^3 + {}^{10}C_8 \left(\frac{1}{2}\right)^8 \left(\frac{1}{2}\right)^2 + {}^{10}C_9 \left(\frac{1}{2}\right)^9 \left(\frac{1}{2}\right)^1 + {}^{10}C_{10} \left(\frac{1}{2}\right)^{10}$$

$$= {}^{10}C_7 \left(\frac{1}{2}\right)^{10} + {}^{10}C_8 \left(\frac{1}{2}\right)^{10} + {}^{10}C_9 \left(\frac{1}{2}\right)^{10} + {}^{10}C_{10} \left(\frac{1}{2}\right)^{10}$$

$$= \frac{1}{2^{10}} \left[{}^{10}C_7 + {}^{10}C_8 + {}^{10}C_9 + {}^{10}C_{10} \right]$$

$$= \frac{1}{2^{10}} \left[\frac{10!}{(10-7)!\,7!} + \frac{10!}{(10-8)!\,8!} + \frac{10!}{(10-9)!\,9!} + \frac{10!}{(10-10)!\,10!} \right]$$

$$= \frac{1}{2^{10}} \left[\frac{10 \times \overset{3}{9} \times \overset{4}{8} \times \cancel{7!}}{\cancel{3} \times \cancel{2} \times 1 \times \cancel{7!}} + \frac{\overset{5}{10} \times 9 \times \cancel{8!}}{\cancel{2} \times 1 \times \cancel{8!}} + \frac{10 \times \cancel{9!}}{1! \times \cancel{9!}} + 1 \right]$$

$$= \frac{1}{2^{10}} \left[120 + 45 + 10 + 1 \right]$$

$$= \frac{176}{2^{10}} = \frac{176}{1024} = 0.1719$$

(ii) Atleast 6 heads

$$P(X \geq 6) = P(X=6) + P(X \geq 7)$$

$$= {}^{10}C_6 \left(\frac{1}{2}\right)^6 \left(\frac{1}{2}\right)^{10-6} + 0.1719$$

$$= {}^{10}C_6 \left(\frac{1}{2}\right)^6 \left(\frac{1}{2}\right)^4 + 0.1719$$

$$= {}^{10}C_6 \left(\frac{1}{2}\right)^{10} + 0.1719$$

$$= \frac{10\times9\times8\times7\times\cancel{6!}}{(10-6)!\,\cancel{6!}}\left(\frac{1}{2}\right)^{10} + 0.1719$$

$$\left(\frac{1}{2}\right)^{10}\left(\frac{1}{2}\right)^{10-10}$$

$$= \frac{10\times\overset{3}{\cancel{9}}\times\overset{4}{\cancel{8}}\times7}{\cancel{4}\times\cancel{3}\times\cancel{2}\times1}\left(\frac{1}{2}\right)^{10} + 0.1719$$

$$= \frac{210}{2^{10}} + 0.1719$$

$$= \frac{210}{1024} + 0.1719$$

$$= 0.205 + 0.1719$$

$$= 0.3769$$

② The probability then John hits a target is $\frac{1}{2}$. He fires 6 times. Find the probability that he hits the target.